Anatomy of a Random Variable

A more in-depth treatment of many of these concepts can be found at: https://github.com/zsunberg/CU-DMUPP-Materials/blob/main/notes/STAT_219_Notes.pdf

Outline

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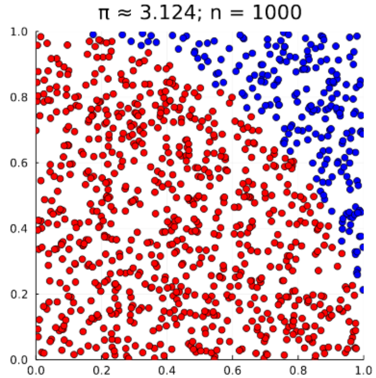

- A Motivating Example: Monte Carlo Integration

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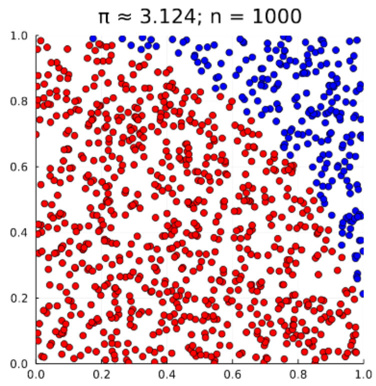
- Rigorous Definitions of a Random Variable



$$X : \Omega \to E$$

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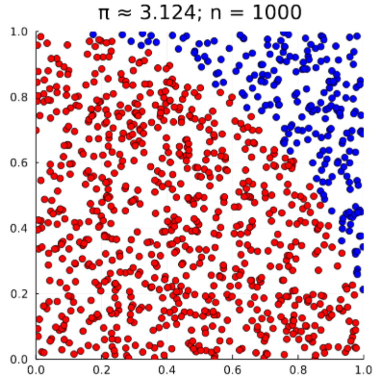
- A Motivating Example: Monte Carlo Integration
- Rigorous Definitions of a Random Variable
- Law of large numbers and the Central Limit Theorem



$$X : \Omega \to E$$

$$\sqrt{n}\left(\bar{X}_n - \mu\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\right)$$

Monte Carlo Integration

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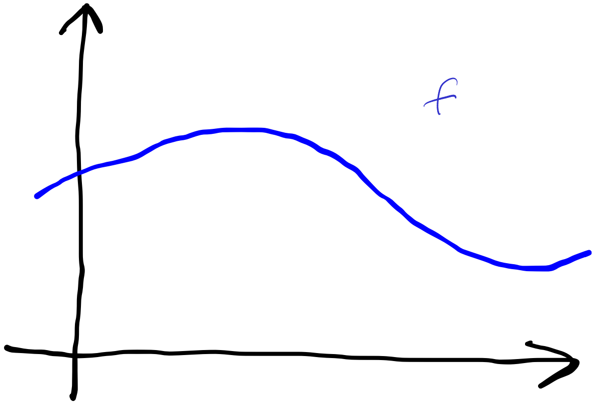

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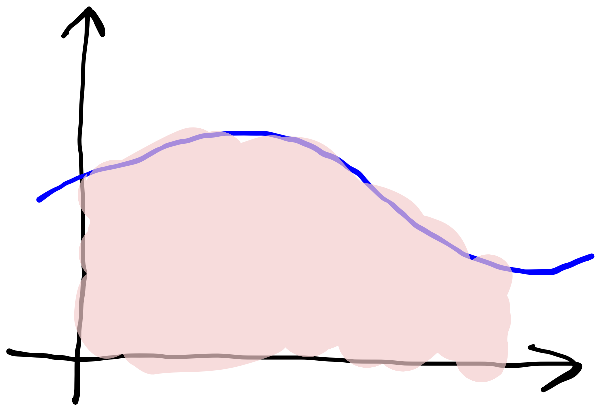

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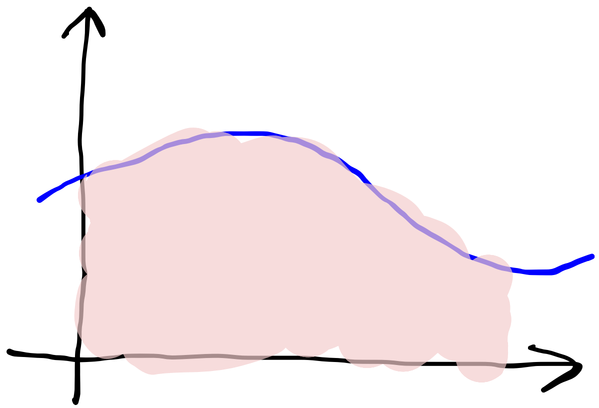
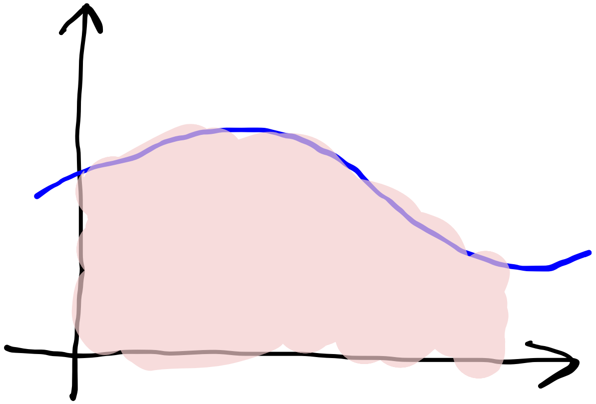
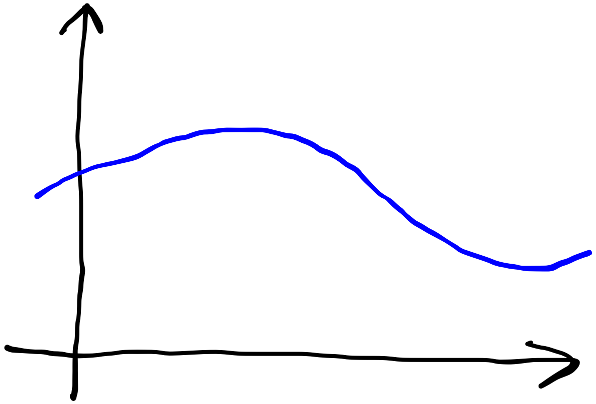


$$I = \int_\Omega f(x)dx$$

$$I = \int_\Omega f(x)\underline{\mu(dx)}$$

Monte Carlo Integration





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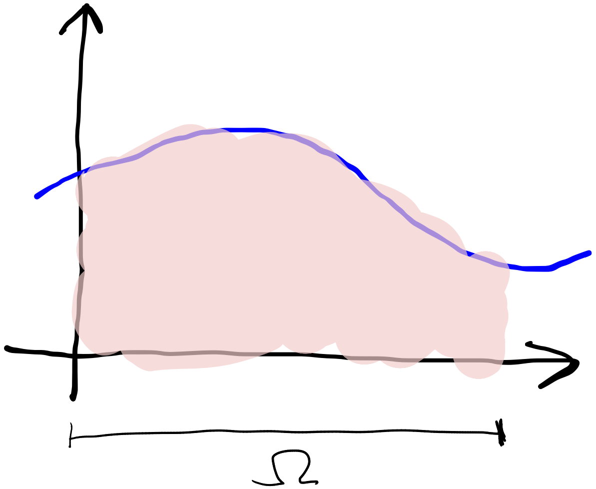

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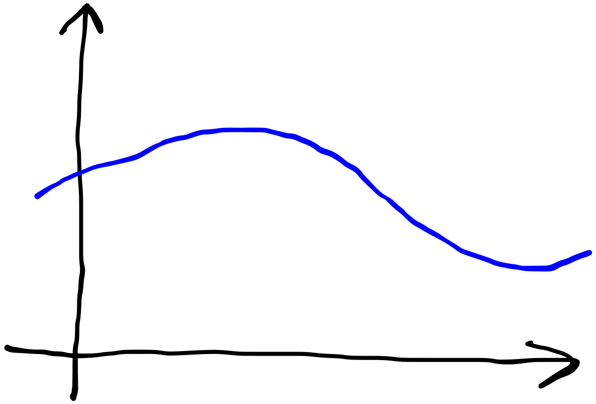
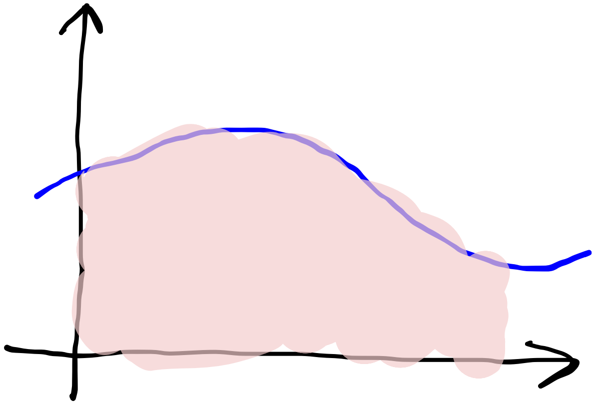
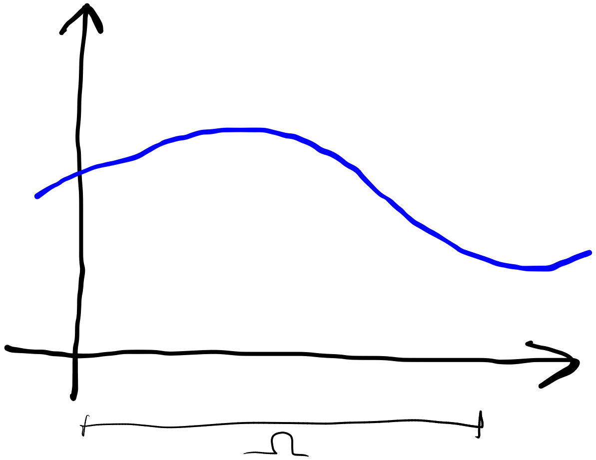


$$X_i \sim \underline{U(\Omega)}$$

Monte Carlo Integration





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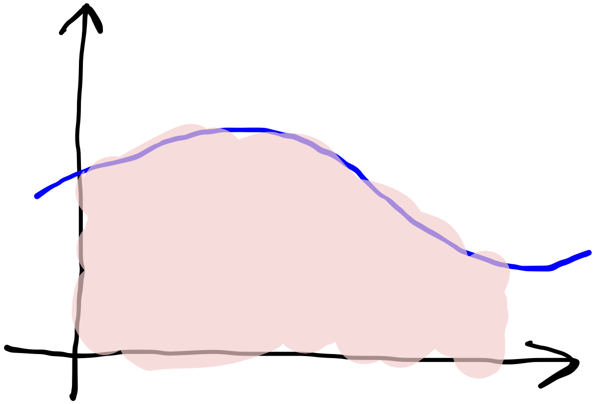

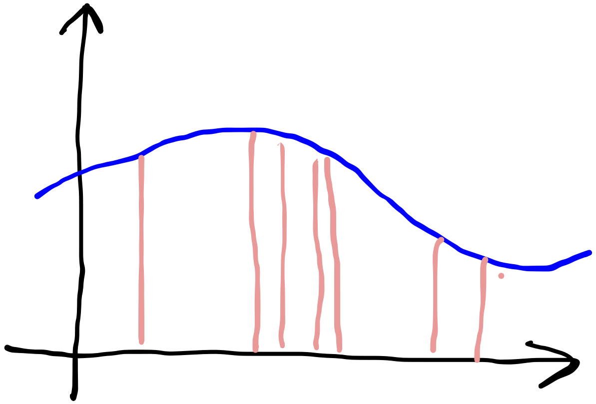

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A useful function: the Indicator

Monte Carlo Integration

Special Case: Expectation

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$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x\, p(x)\, dx$$

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In what sense does this converge as $N \to \infty$? How accurate is it?

Random Variables

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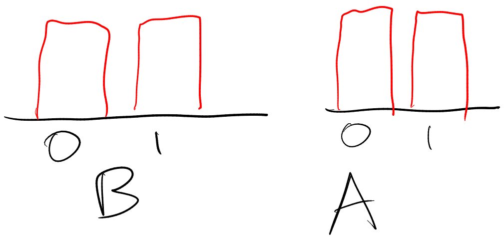

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Given a **probability space** $(\Omega, \mathcal{F}, P)$, and a **measurable space** $(E, \mathcal{E})$, an E-valued **random variable is a measurable function** $X : \Omega \to E$.

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What is this function?

What is $\omega \in \Omega$?

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Example: Coin World

$$\Omega = \{H, T\}$$

$$E = [0, 1]$$

$$X(\omega) = \underline{\mathbf{1}_{\{H\}}}(\omega)$$

$$\mathbf{1}_A(x) \equiv \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{o.w.} \end{cases}$$

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$$\Omega = \{H, T\}^n$$

$$E = [0, 1]$$

$$X_i(\omega) = \mathbf{1}_{\{H\}}(\omega_i)$$

X_1

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Example: Coin World

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Example: Many coins

$$\Omega = \{H, T\}^\infty$$

$$E = [0, 1]$$

$$X_i(\omega) = \mathbf{1}_{\{H\}}(\omega_i)$$

$$Y(\omega) = \sum_{i=1}^{1000} X_i(\omega) = \sum_{i=1}^{1000} \mathbf{1}_{\{H\}}(\omega_i)$$

$$Z(\omega) = \sum_{i=2}^{1001} X_i(\omega)$$

What is $\omega \in \Omega$?

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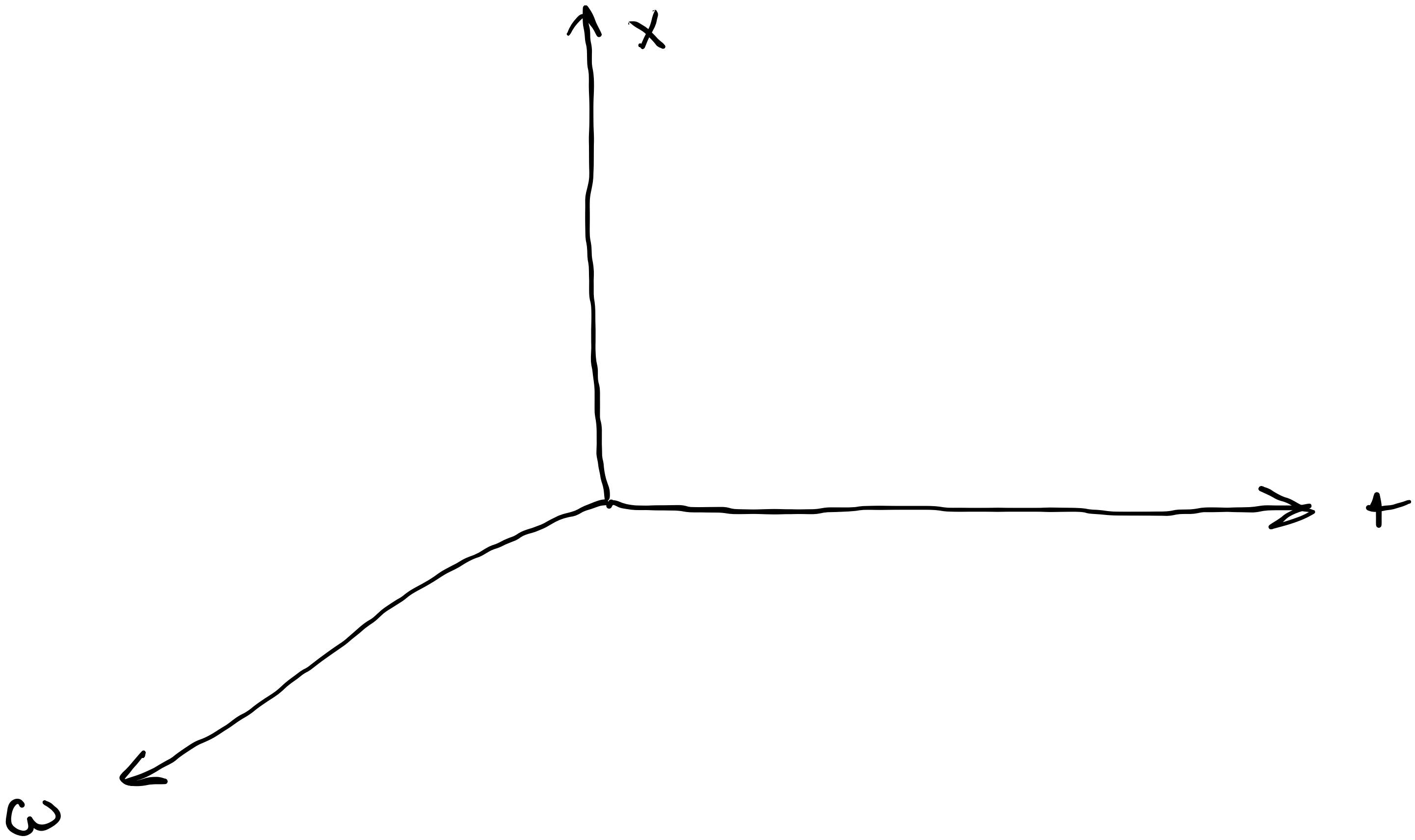
$x(t)$



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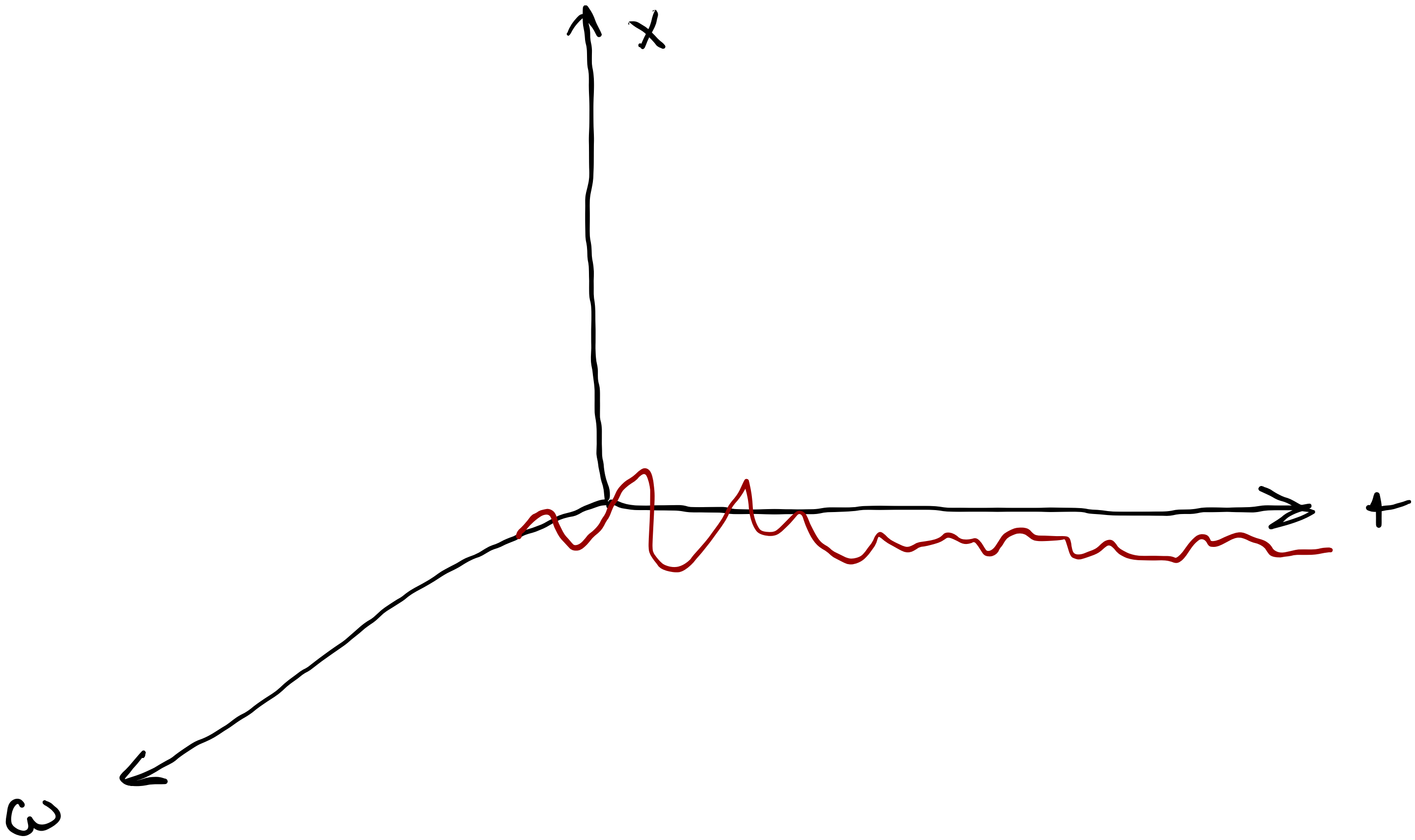

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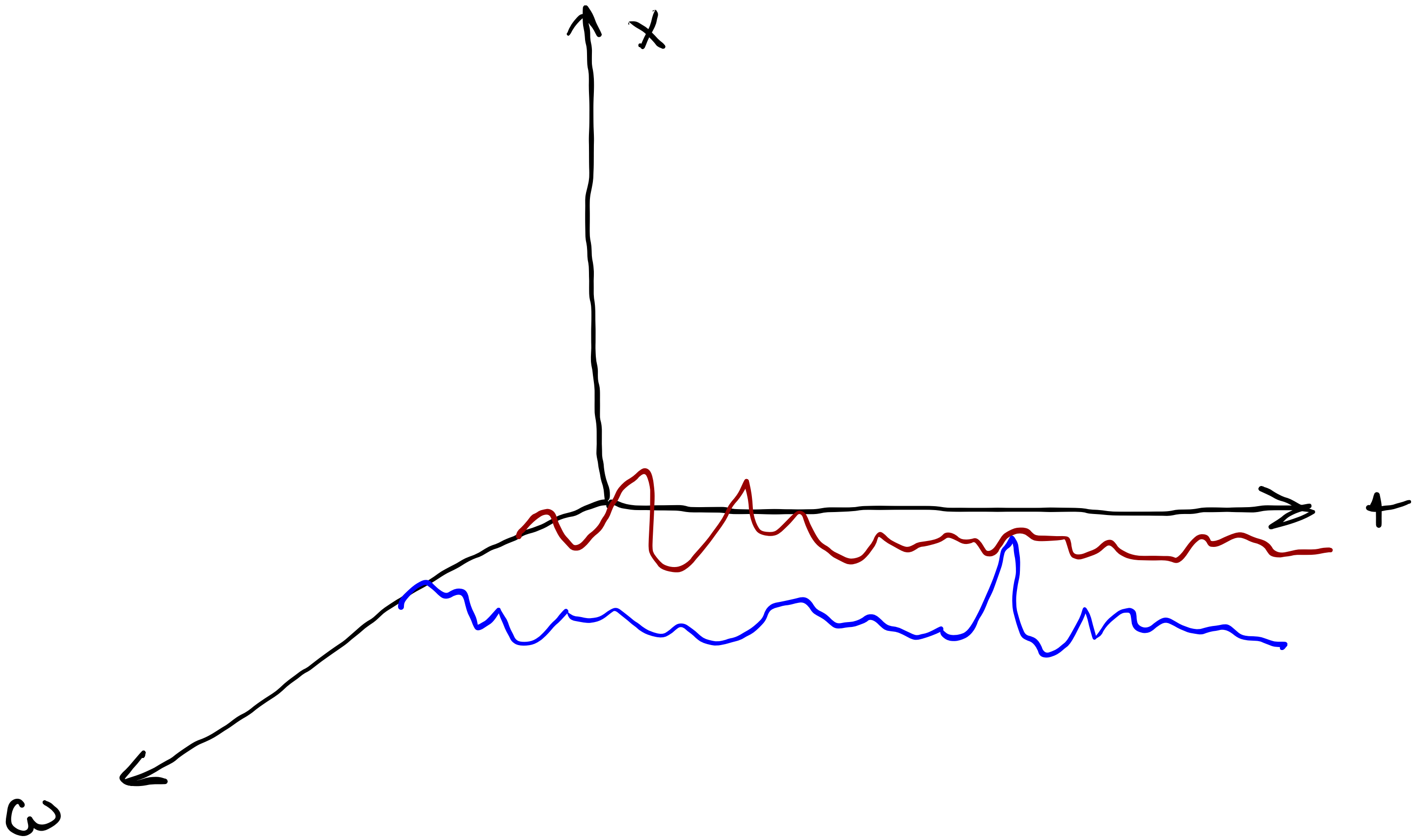

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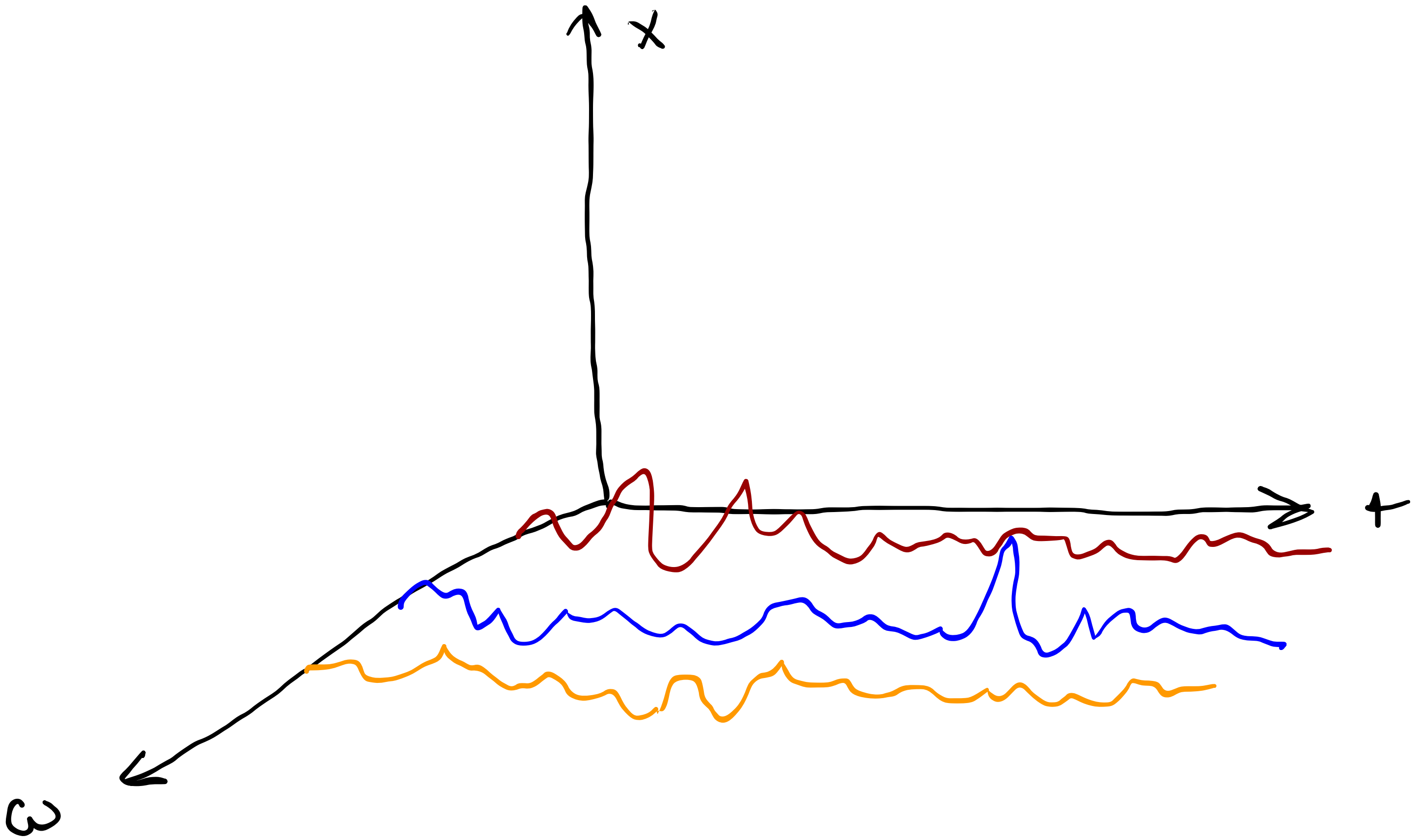

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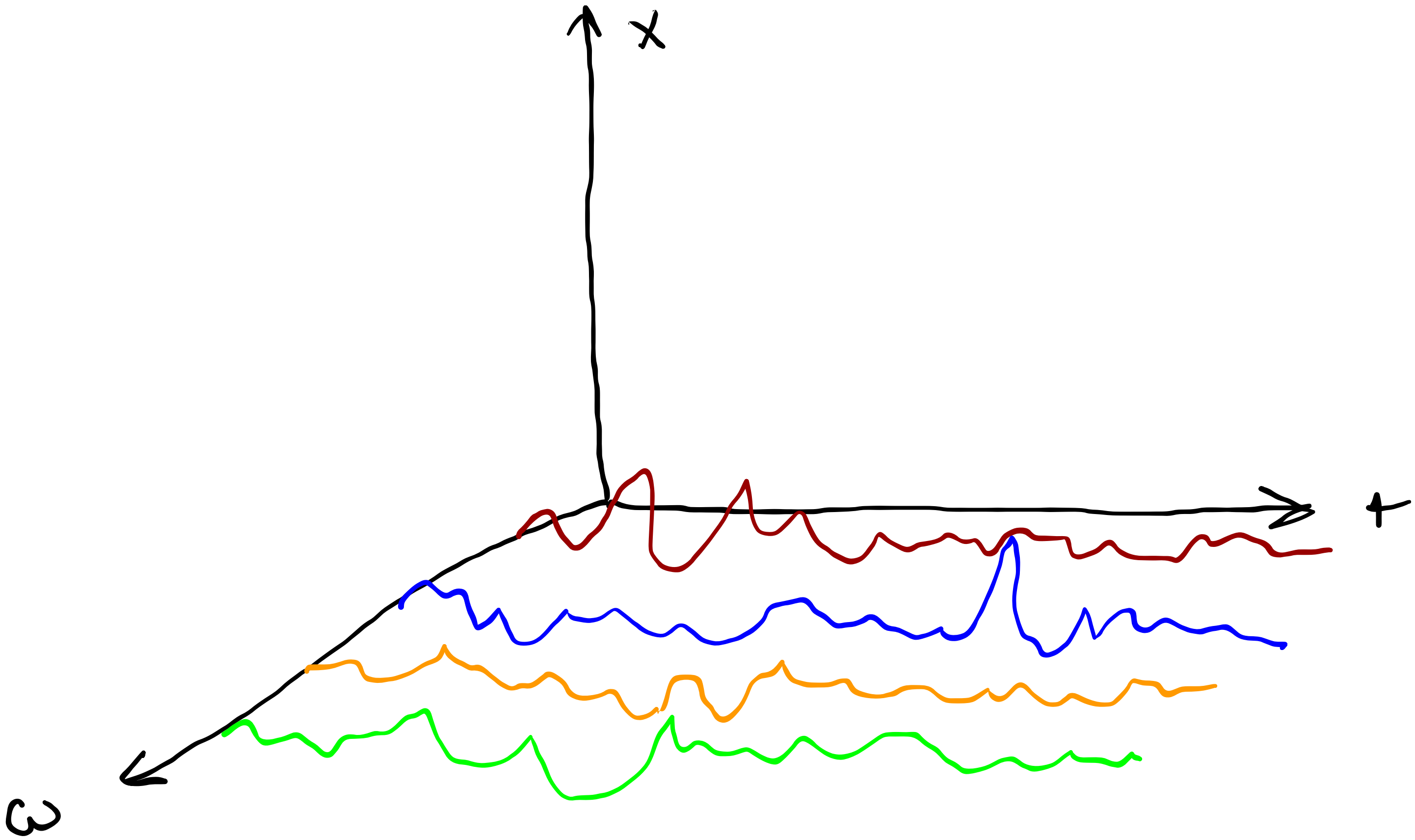

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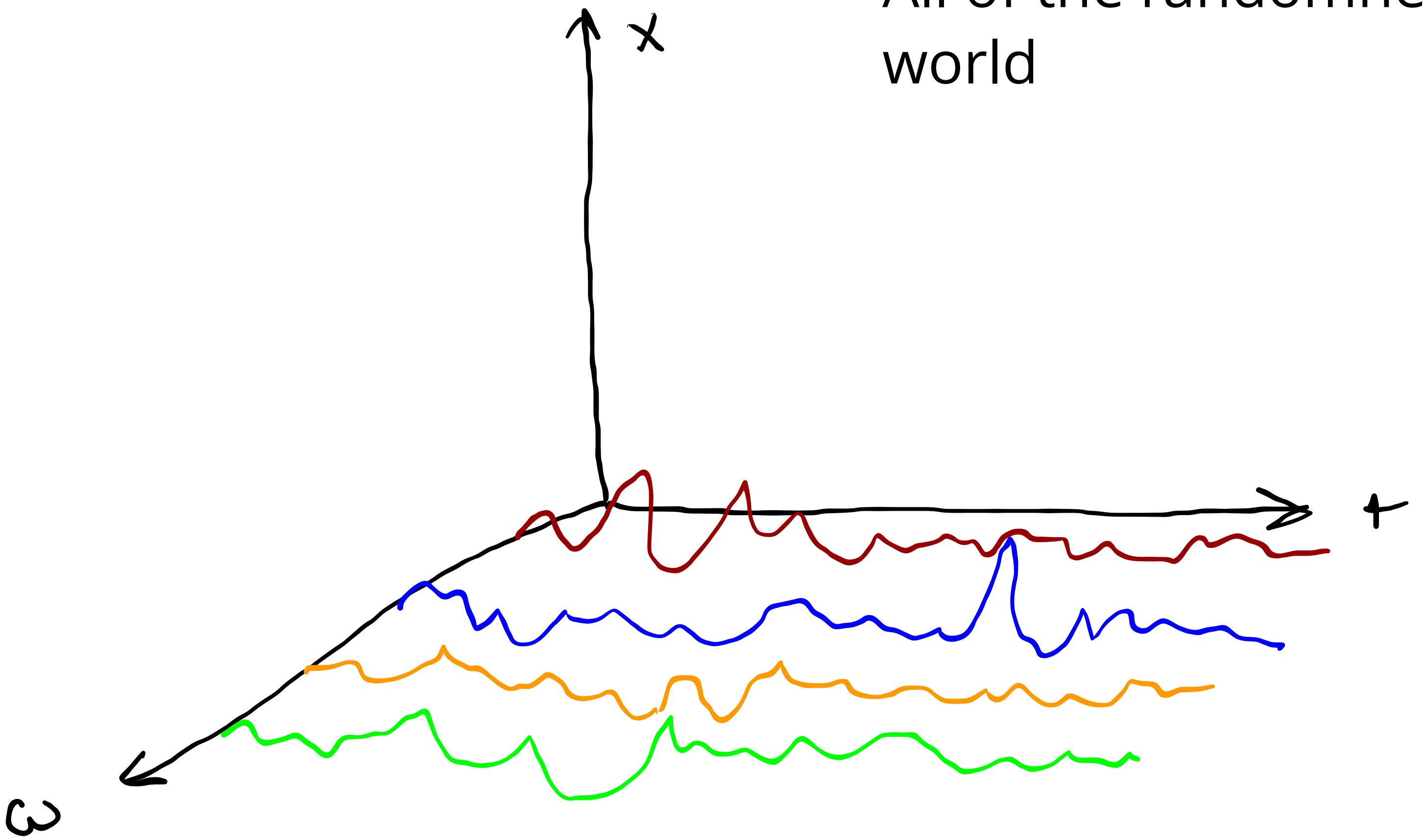

- All of the randomness in the world

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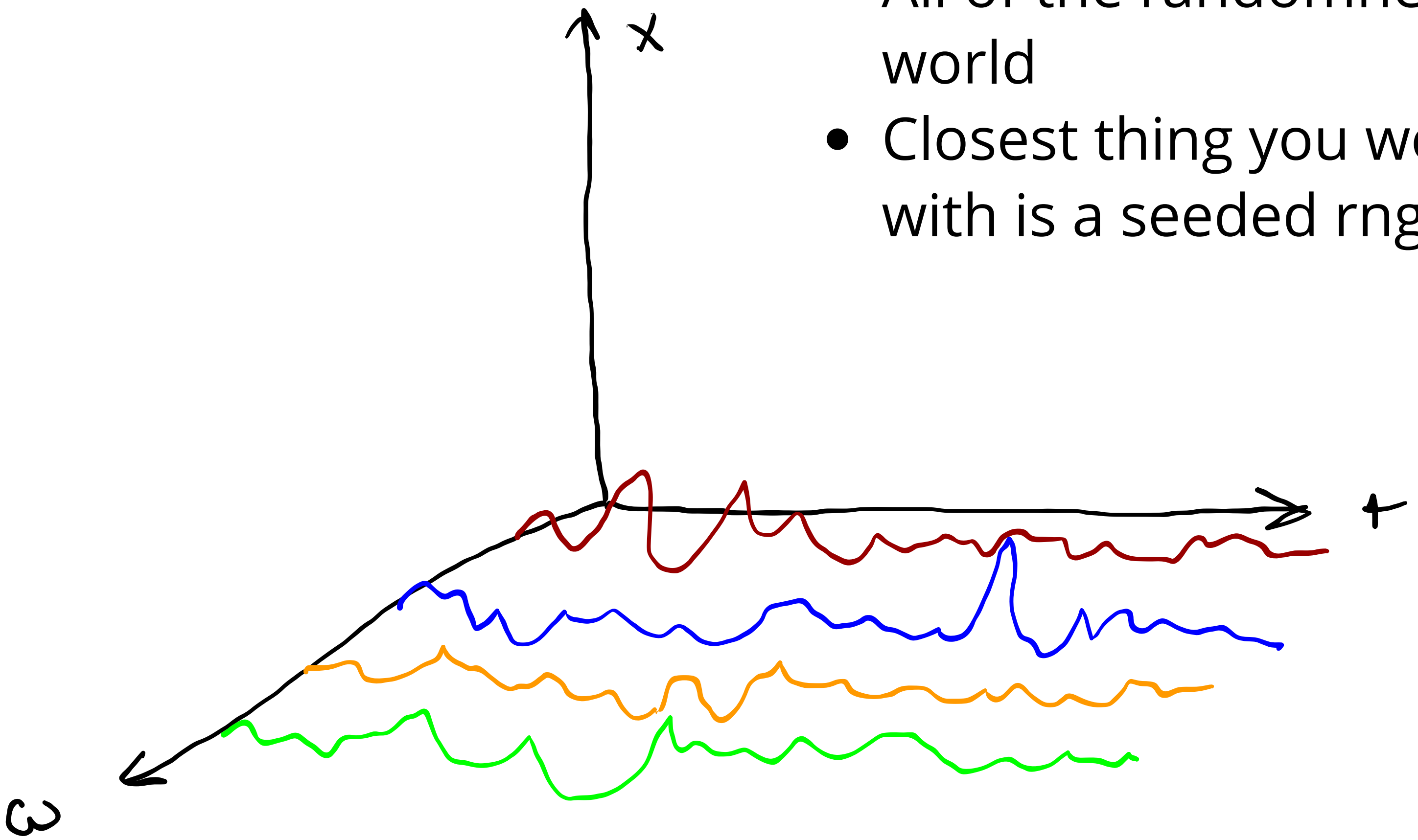


- All of the randomness in the world
- Closest thing you work regularly with is a seeded rng

What is $\omega \in \Omega$?

$\omega \in \Omega = 0 \ldots. 10000$

$E = \mathbb{R}^N$

$X(\omega)$ defined algorithmically as follows:

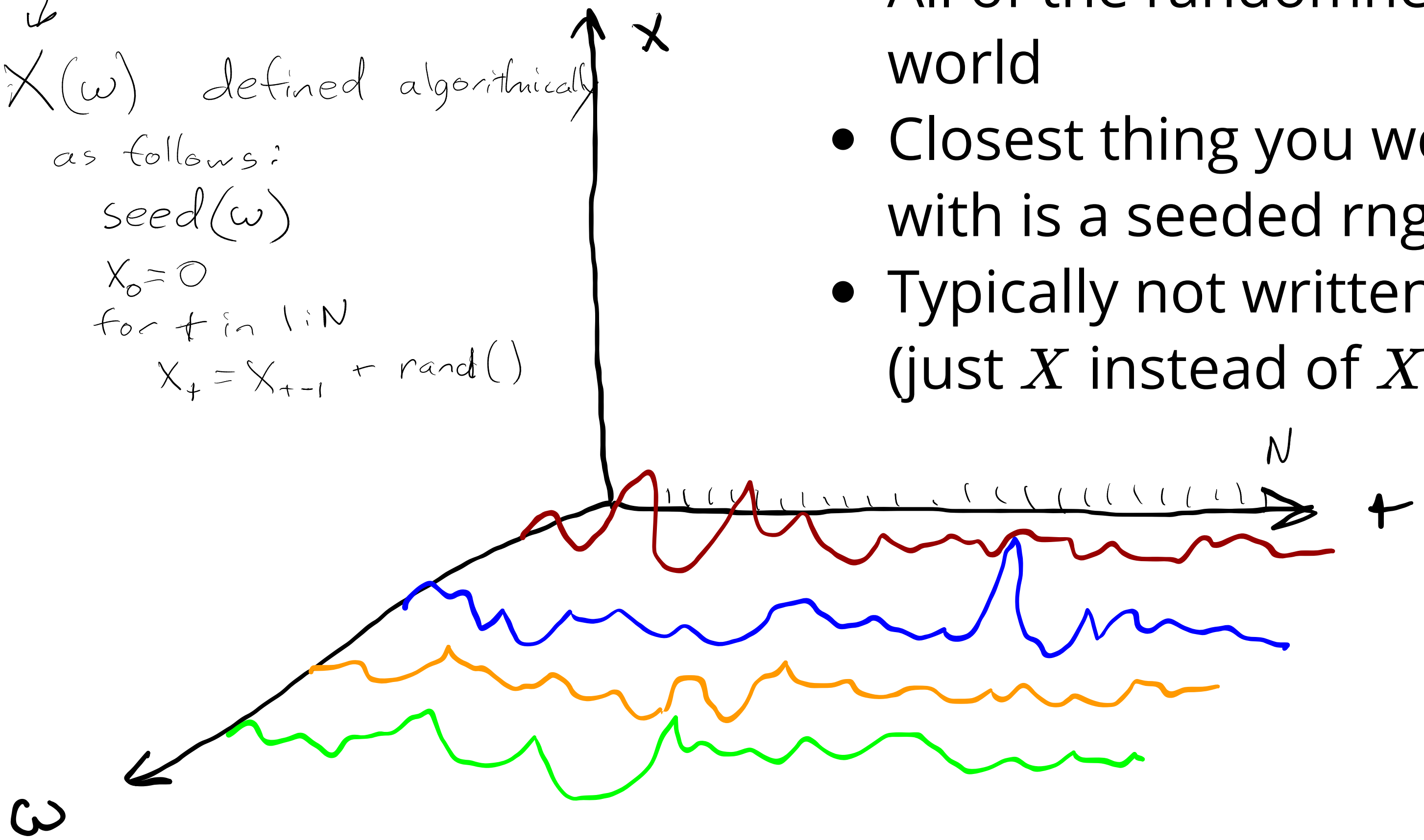
seed(ω)

$X_0 = 0$

for t in $1:N$

$X_t = X_{t-1} + \text{rand}()$

- All of the randomness in the world
- Closest thing you work regularly with is a seeded rng
- Typically not written expressly (just X instead of $X(\omega)$)



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$\Omega = \{1, 2, 3\}$

$2^{\Omega} = \{\{1\}, \{2\}, \{3\}, \varnothing, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

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    - If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$ (where $A^c = \Omega \backslash \overset{A}{\cancel{B}}$)
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complement

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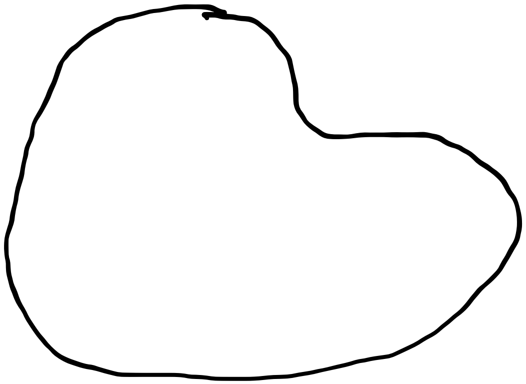
Ω



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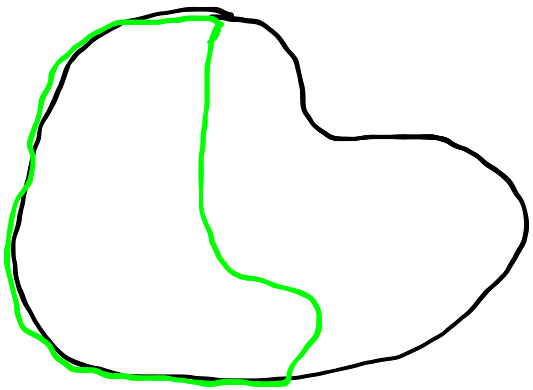

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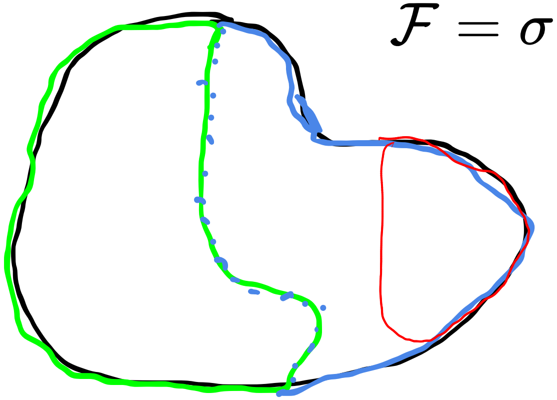

$\mathcal{F} = \sigma(\{\ \}) = \{\underline{\Omega}, \ , \ , \underline{\emptyset}\}$

$\sigma(\{B, D\}) =$
$\{\Omega, B, D, B, D, \phi, \cdot\ \cdot, (\cdot\ \cdot)^c\}$

σ-algebra Examples

If $\Omega = \{1, 2, 3\}$, what is $\sigma(\{1\})$?

$\{\{1\}, \{2, 3\}, \{1, 2, 3\}, \phi\}$

If $\Omega = \{1, 2, 3\}$, what is $\sigma(\{1, 2\})$?

$\{\{1, 2, 3\}, \phi, \{1, 2\}, \{3\}\}$

Borel Sigma Algebra

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The Borel σ-algebra for a topological space Ω is the σ-field generated by all open sets in Ω.

The Borel σ-field on $\mathbb{R}$ is $\mathcal{B} = \sigma(\{(a, b) : a, b \in \mathbb{R}\})$

open intervals

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$(1,\infty) = \bigcup_{i=1}^{\infty} (1,i)$

- Is $(1,2)$ in $\mathcal{B}$?
- Is $(1,\infty)$ in $\mathcal{B}$? Yes
- Is $[1,2]$ in $\mathcal{B}$? Yes

$(-\infty,1) \cup (2,\infty)$

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- Is $(1, 2)$ in $\mathcal{B}$?
- Is $(1, \infty)$ in $\mathcal{B}$?
- Is $[1, 2]$ in $\mathcal{B}$?
- Is 1 in $\mathcal{B}$?

$\left((-\infty, 1) \cup (1, \infty)\right)^c$

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2. $P(\Omega) = 1$.

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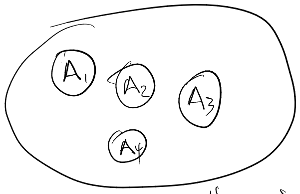
A probability measure P is a function $P : \mathcal{F} \to [0, 1]$ having the following properties:



$P\left(\bigcup_{i=1}^{4} A_i\right) = \sum_{i=1}^{4} P(A_i)$

1. $0 \leq P(A) \leq 1 \quad \forall\, A \in \mathcal{F}$.
2. $P(\Omega) = 1$.
3. (Countable additivity) $P(A) = \sum_{n=1}^{\infty} P(A_n)$ whenever $A = \cup_{n=1}^{\infty} A_n$ is a countable union of disjoint sets $A_n \in \mathcal{F}$

Random Variables

Given a **probability space** $(\Omega, \mathcal{F}, P)$, and a **measurable space** $(E, \mathcal{E})$, an E-valued **random variable is a measurable function** $X : \Omega \to E$.

Random Variables

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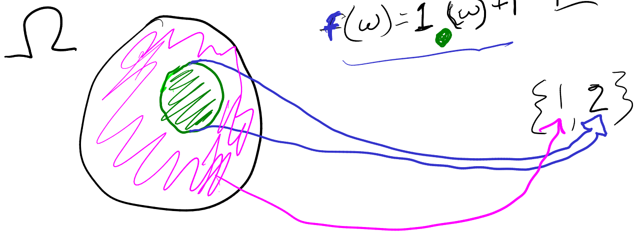
A function $f : \Omega \to E$ is measurable if for every $A \in \mathcal{E}$, the pre-image of A under f is in $\mathcal{F}$. That is, for all $A \in \mathcal{E}$

$$f^{-1}(A) \in \mathcal{F}$$

$\mathcal{F} = \sigma(\bullet)$

$f(\omega) = 1_{\bullet}(\omega) + 1$

E



$X(\omega) = f(\omega)$

$P(X=1) = P(\bullet)$

Random Variables

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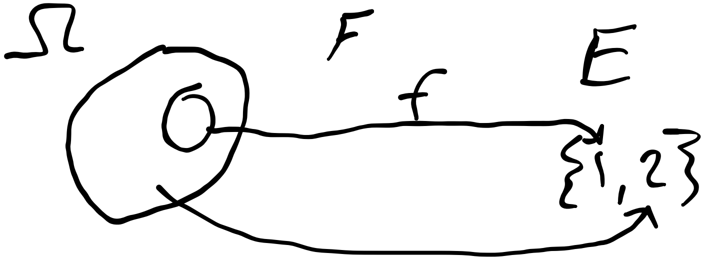

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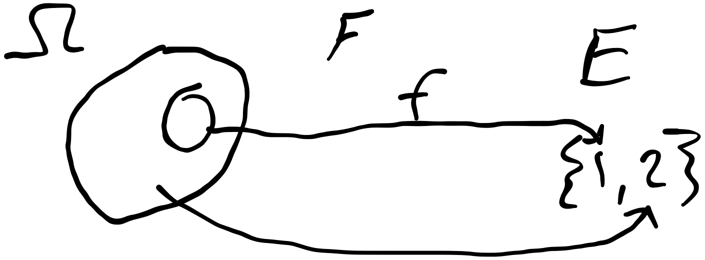

Are there functions that are not Borel-measurable?

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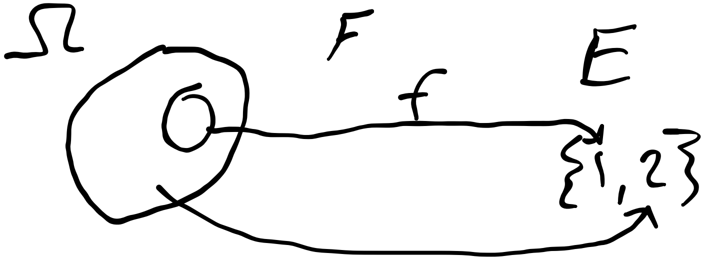

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Are there functions that are not Borel-measurable?

Yes! Example: $\mathbf{1}_V$ where V is a Vitali Set

Advantages over pdf definition

- More sophisticated convergence concepts
- Better way of thinking about related random variables (personally, I think)

$\Omega = \{H, T\}$ $\quad \mathcal{F} = \sigma(\{H\})$

$P(\{H\}) = \frac{1}{2}$

$$\begin{tikzcd} & \omega \arrow[dl] \arrow[dr] & \\ A \arrow[rr, leftrightarrow] & & B \end{tikzcd}$$

$A(\omega) = \mathbb{1}_{\{H\}}(\omega)$

$B(\omega) = \mathbb{1}_{\{T\}}(\omega)$

Break

Exercise 1.2.5. *Let* $\Omega = \{1, 2, 3\}$. *Find a* σ*-field* $\mathcal{F}$ *such that* $(\Omega, \mathcal{F})$ *is a measurable space, and a mapping* X *from* Ω *to* $\mathbb{R}$, *such that* X *is not a random variable on* $(\Omega, \mathcal{F})$.

Hint:

1. Use $\sigma(\{1\})$ (what is it?)
2. What function has a preimage not in $\sigma(\{1\})$?

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Hint:

1. Use $\sigma(\{1\})$ (what is it?)

2. What function has a preimage not in $\sigma(\{1\})$?

$$\mathcal{F} = \{\Omega, \emptyset, \{1\}, \{2, 3\}\}$$

$$X = \mathbf{1}_{\{1,2\}}$$

Convergence

Convergence

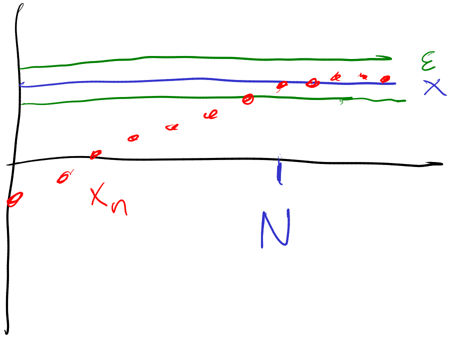
Review: For a (deterministic) sequence $\{x_n\}$, we say

$$\lim_{n\to\infty} x_n = x$$

or

$$x_n \to x$$

if, for every $\epsilon > 0$, there exists an N such that $|x_n - x| < \epsilon$ for all $n > N$.



Convergence

In what senses can we talk about a sequence of random variables converging?

- Sure ("pointwise")
- Almost Sure
- In Probability
- Weak ("in distribution"/"in law")

When are two R.V.'s the same?

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We say that X is *almost surely* the same as Y if $P(\{\omega : X(\omega) \neq Y(\omega)\}) = 0.$

This is denoted $X \overset{a.s.}{=} Y$ and the terms *almost everywhere* (a.e.) and *with probability 1* (w.p.1) mean the same thing.

Sure Convergence

$$X_n(\omega) \to X(\omega) \quad \forall\, \omega \in \Omega$$

Almost Sure Convergence

Almost Sure Convergence

$X_n \overset{a.s.}{\to} X$ if there exists $A \in \mathcal{F}$ with $P(A) = 1$ such that $X_n(\omega) \to X(\omega)$ for each fixed $\omega \in A$.

Almost Sure Convergence

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Does sure convergence imply almost sure convergence?

Convergence in Probability

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$X_n \to_p X$ if $P(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) \to 0$ for any fixed $\epsilon > 0$.

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Yes.

Convergence in Probability

Does $X_n \rightarrow_p X$ imply $X_n \overset{a.s}{\rightarrow} X$?

Convergence in Probability

Does $X_n \to_p X$ imply $X_n \stackrel{a.s}{\to} X$?

No.

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PROOF. Consider the probability space $\Omega = (0,1)$, with Borel σ-field and the Uniform probability measure U of Example 1.1.11. Suffices to construct an example of $X_n \to_p 0$ such that fixing each $\omega \in (0,1)$, we have that $X_n(\omega) = 1$ for infinitely many values of n. For example, this is the case when $X_n(\omega) = \mathbf{1}_{[t_n, t_n+s_n]}(\omega)$ with $s_n \downarrow 0$ as $n \to \infty$ slowly enough and $t_n \in [0, 1-s_n]$ are such that any $\omega \in [0,1]$ is in infinitely many intervals $[t_n, t_n + s_n]$. The latter property applies if $t_n = (i-1)/k$ and $s_n = 1/k$ when $n = k(k-1)/2 + i$, $i = 1, 2, \ldots, k$ and $k = 1, 2, \ldots$ (plot the intervals $[t_n, t_n + s_n]$ to convince yourself). ∎

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But there exists a subsequence n_k such that $X_{n_k} \overset{a.s.}{\to} X$.

Weak Convergence

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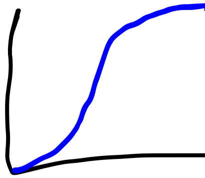
$F_X(x) = P(X \leq x)$



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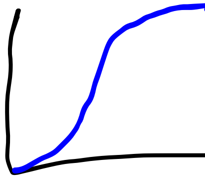

$X_n \xrightarrow{D} X$ if $F_{X_n}(\alpha) \to F_X(\alpha)$ for each fixed α that is a continuity point of F_X.

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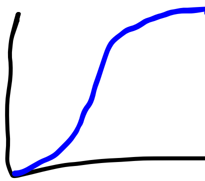

$X_n \xrightarrow{D} X$ if $F_{X_n}(\alpha) \to F_X(\alpha)$ for each fixed α that is a continuity point of F_X.

"Weak convergence", "convergence in distribution", and "convergence in law" all mean the same thing.

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Convergence

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Stronger

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- Almost Sure
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Implies

Convergence of MC integration

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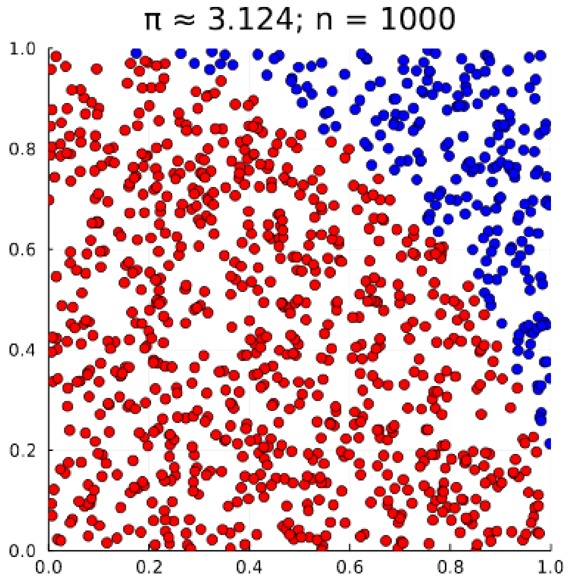
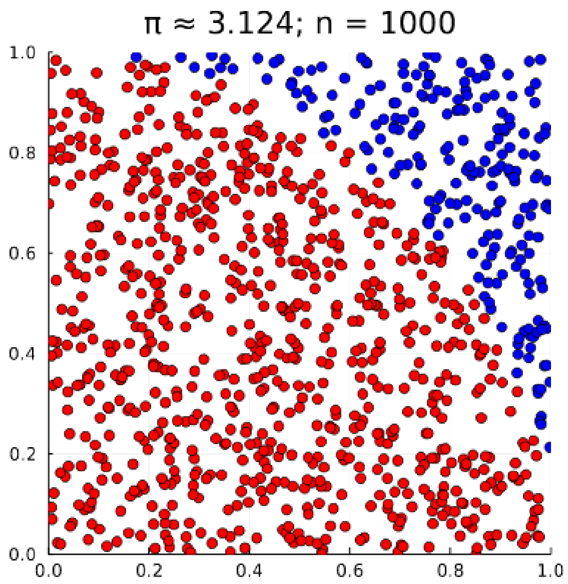
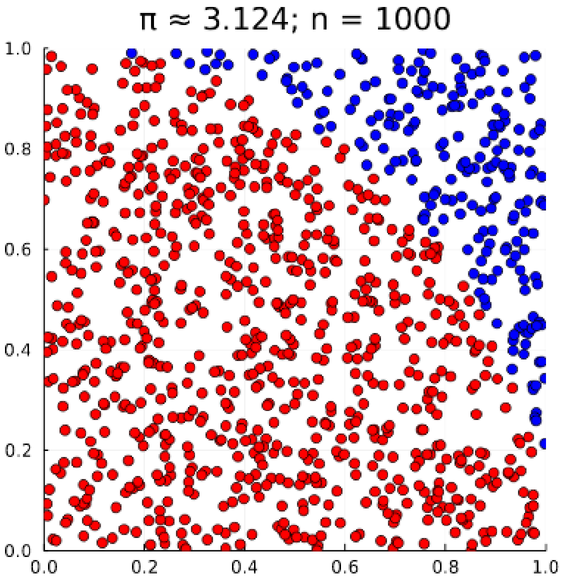


Convergence of MC integration



Let X_i be independent, identically distributed random variables with mean μ, and $Q_N \equiv \frac{1}{N} \sum_{i=1}^{N} X_i$.

Convergence of MC integration



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$$Q_N \xrightarrow{?} \mu?$$

Convergence of MC integration

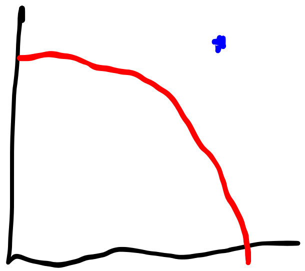
$$Q_N \to \mu \text{ (sure)?}$$

$$Q_N \overset{a.s.}{\to} \mu?$$

$$Q_N \to_p \mu?$$

$$Q_N \overset{D}{\to} \mu?$$

Convergence of MC integration



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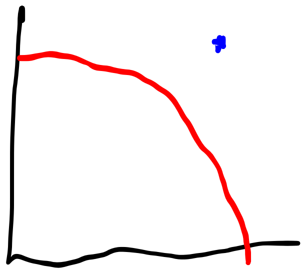
Convergence of MC integration

$\exists \omega \in \Omega$ where you always sample the same point.



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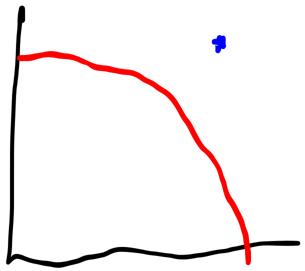

$$Q_N \to \mu \text{ (sure)?}$$

$$Q_N \stackrel{a.s.}{\to} \mu?$$

Probability that there are enough measurements off in one direction to keep $|Q_N - \mu| > \epsilon$ decays with more samples.

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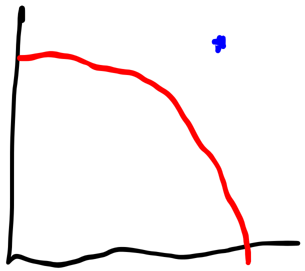

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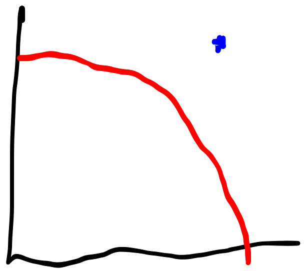

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Weak law of large numbers

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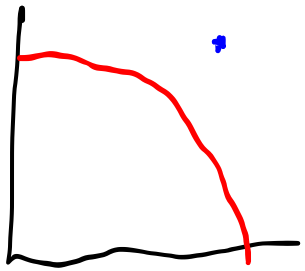

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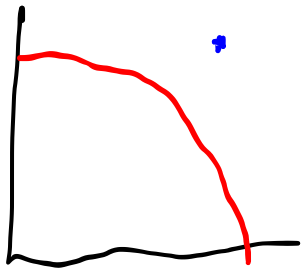

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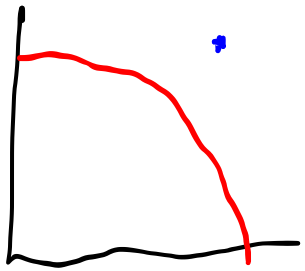

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Convergence *Rate* of M.C. Integration

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How do you quantify $|Q_N - \mu|$?

Convergence *Rate* of M.C. Integration

How do you quantify $|Q_N - \mu|$?

Run M sets of N simulations and plot a histogram of Q_N^j for $j \in \{1, \dots, M\}$.

Central Limit Theorem

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Lindeberg-Levy CLT: If $\mathrm{Var}[X_i] = \sigma^2 < \infty$, then

$$\sqrt{N}(Q_N - \mu) \xrightarrow{D} \mathcal{N}(0, \sigma)$$

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After many samples Q_N starts to look distributed like $\mathcal{N}(\mu, \frac{\sigma}{\sqrt{N}})$

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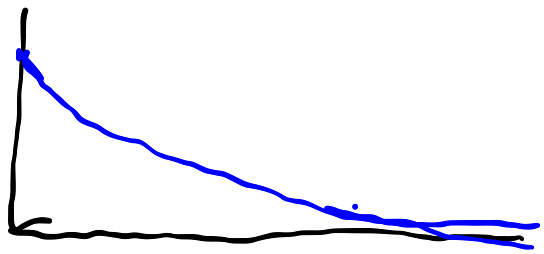

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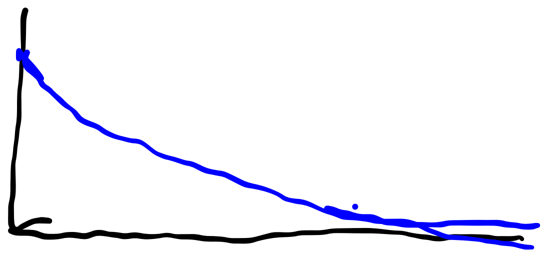

2. You can estimate the "standard error" with

$$SE = \frac{s}{\sqrt{N}}$$

where s is the sample standard deviation.